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Roll No

EE/EX-221-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS) **Electrical Measurements and Instrumentation**

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume any missing data.

1. a) Define the terms
i) Accuracy ii) Precision
iii) Static sensitivity iv) Reliability
b) What do you mean by calibration curve? Explain loading effects due to shunt connected and series connected instruments.
2. a) Explain the construction and working of D'Arsonval Galvanometer.
b) Explain why PMMC instruments are the most widely used instruments. Discuss their advantages and disadvantages.
3. a) Explain with the help of block diagram the working of Digital voltmeter.
b) Write down the difference between current transformer and potential transformer. Also explain their working.
4. a) Describe the constructional details and working of an electro-dynamometer type of wattmeter. Discuss the main sources of errors in electro-dynamometer type instruments.

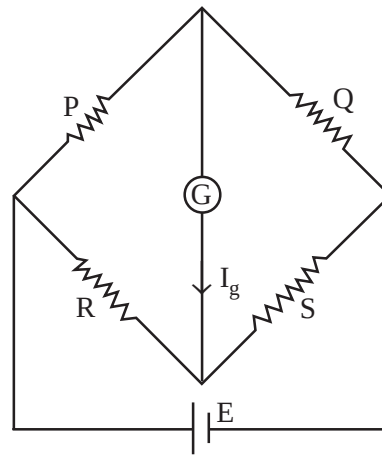
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- b) Explain the measurement of power using
 - i) Three wattmeter method
 - ii) CTs and PTs
- 5. a) Explain the construction and operation of single phase and three phase electronic energy meter.
- b) What is Phantom loading? Explain with example. Also describe the testing by phantom loading for energy meters.
- 6. a) Explain the working of Weston type frequency meter.

- b) A Wheatstone bridge is shown in figure. The values of resistance are $P = 1\text{k}\Omega$, $R = 1\text{k}\Omega$, $S = 5\text{k}\Omega$, $G = 100\Omega$. The thevenin source generator voltage $E_0 = 24\text{mV}$ and the galvanometer current is $13.6\mu\text{A}$. Calculate the value of Q .



- 7. a) Describe the circuit diagram of a series type ohm meter. Explain how is it designed. Why are series type ohm meter preferred over shunt type ohm meters?
- b) What is a B-H curve in terms of magnetic material? Write down its characteristics?
- 8. Write short notes on (any two):
 - a) Lloyd fisher square method
 - b) Economics of measurement system
 - c) Kelvin's double bridge
 - d) Maximum demand meter
